HcZZ ctag2d + JES/JER — Results

Production: combine_run3_100150_ctag2d_v8_jecshifts

hplusc_mva_4class_ctag2d_scored_v8 + hplusc_mva_4class_ctag2d_jecshifts_scored_v2, [100,150] GeV, no-ZX (MC-only), 15 bins (quantile + surgical rebin "2-4,7-10"), 22 nuisances. Re-verified 2026-09-09 directly against the stored ROOT/JSON.

Yields [100,150] GeV

Process	Rate
Signal	0.053130
ggZZ	2.941801
qqZZ	36.318223
Other_Higgs	55.077609

S/√B = 0.0055

Final limits (AsymptoticLimits, --run blind)

Quantile	r	κc
2.5%	196.3438	36.70
16%	273.8452	50.15
50% (median)	**412.0000**	**74.11**
84%	648.4839	115.09
97.5%	1004.9548	176.84

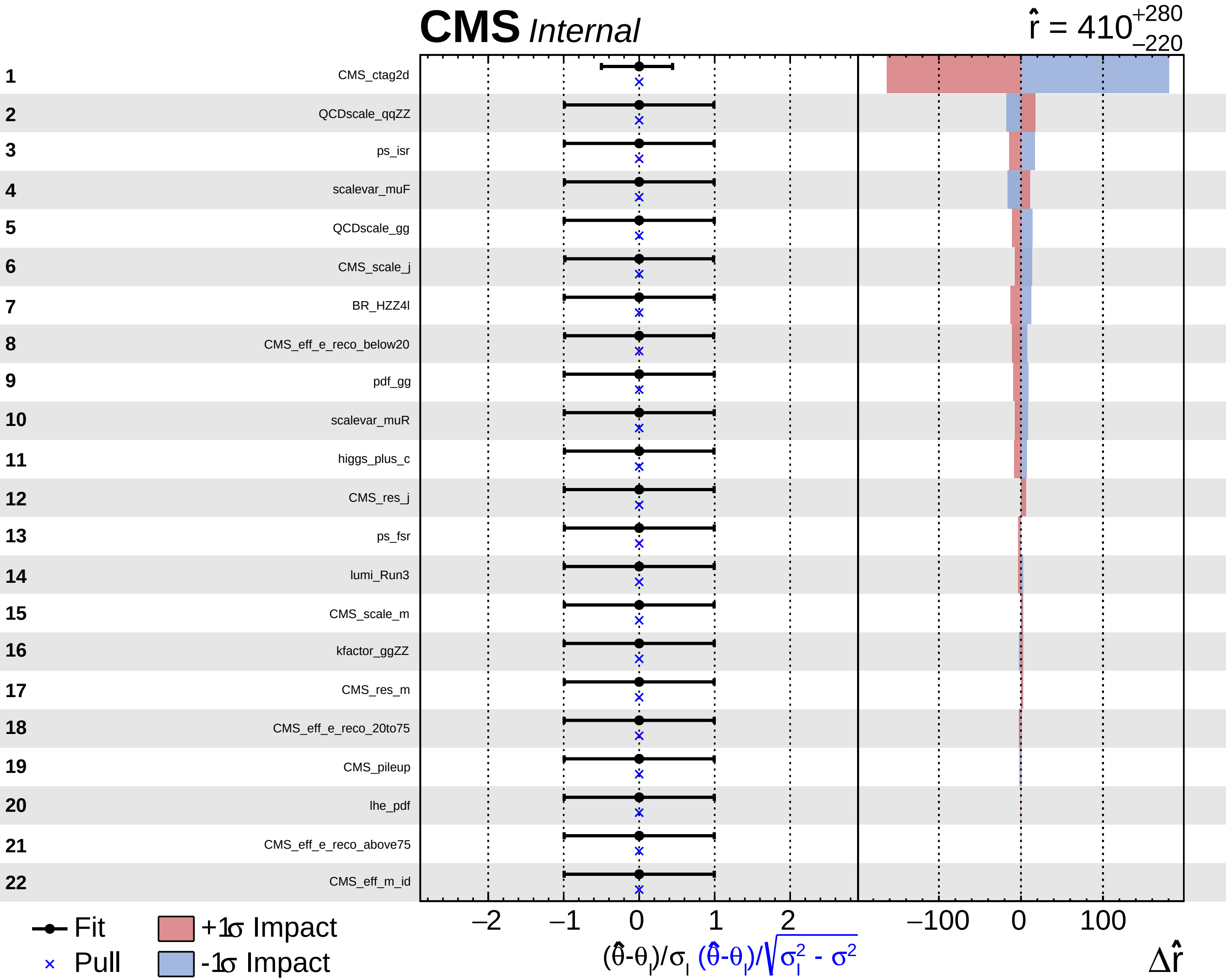
Without JES/JER (same production, JES/JER rows dropped): r=411.0/κc=73.93 — the 4 added nuisances shift the headline median ~0.2%.

JES/JER's 4 rows (CMS_scale_j/CMS_res_j/CMS_scale_m/CMS_res_m) rank #6/#12/#15/#17 by impact — individually bigger than several already-established nuisances (BR_HZZ4l, pdf_gg, scalevar_muR, higgs_plus_c), even though adding all 4 barely moves the headline median. Likely explanation: CMS_ctag2d already dominates the fit (~10x the #2 nuisance), and JES/JER's shape is largely degenerate with/absorbed by it — plausible since JES shifts change which jets get c-tagged. Not independently verified via a correlation matrix.

Systematics: type, assigned value, and impact

Ranked by impact_r (combineTool.py -M Impacts, --expect-signal 412.0). lnN rows show the assigned size per affected process; shape rows have no single size, so the value column shows the post-fit constraint instead (all ≈ prefit ±1σ here — expected for a blind Asimov fit, not a sign of anything wrong).

#	Systematic	Type	Value (lnN size / pull)	Impact
1	**CMS_ctag2d**	**shape**	**pull -0.00 (-0.501/+0.443)**	**180.46**
2	QCDscale_qqZZ	lnN	1.04 (qqZZ)	17.57
3	ps_isr	shape	pull +0.00 (-0.992/+0.993)	17.19
4	scalevar_muF	shape	pull +0.00 (-0.990/+0.993)	16.15
5	QCDscale_gg	lnN	1.039 (ggZZ, Signal)	14.32
6	CMS_scale_j	shape (JES/JER)	pull -0.00 (-0.985/+0.985)	13.64
7	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	12.64
8	CMS_eff_e_reco_below20	shape	pull -0.00 (-0.987/+0.987)	10.63
9	pdf_gg	lnN	1.032 (ggZZ, Signal)	9.34
10	scalevar_muR	shape	pull +0.00 (-0.993/+0.994)	8.84
11	higgs_plus_c	shape	pull -0.00 (-0.990/+0.991)	8.50
12	CMS_res_j	shape (JES/JER)	pull -0.00 (-0.993/+0.992)	6.01
13	ps_fsr	shape	pull +0.00 (-0.991/+0.994)	3.43
14	lumi_Run3	lnN	1.014 (all processes)	3.30
15	CMS_scale_m	shape (JES/JER)	pull +0.00 (-0.992/+0.992)	2.67
16	kfactor_ggZZ	lnN	1.10 (ggZZ)	2.63
17	CMS_res_m	shape (JES/JER)	pull +0.00 (-0.992/+0.992)	2.50
18	CMS_eff_e_reco_20to75	shape	pull -0.00 (-0.994/+0.994)	2.23
19	CMS_pileup	shape	pull +0.00 (-0.994/+0.993)	1.79
20	lhe_pdf	shape	pull -0.00 (-0.994/+0.994)	0.35
21	CMS_eff_e_reco_above75	shape	pull +0.00 (-0.994/+0.994)	0.12
22	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.02



CMS Internal
$\hat{r} = 410^{+280}_{-220}$
1 CMS_ctag2d
2 QCDscale_qqZZ
3 ps_isr
4 scalevar_muF
5 QCDscale_gg
6 CMS_scale_j
7 BR_HZZ4l
8 CMS_eff_e_reco_below20
9 pdf_gg
10 scalevar_muR
11 higgs_plus_c
12 CMS_res_j
13 ps_fsr
14 lumi_Run3
15 CMS_scale_m
16 kfactor_ggZZ
17 CMS_res_m
18 CMS_eff_e_reco_20to75
19 CMS_pileup
20 lhe_pdf
21 CMS_eff_e_reco_above75
22 CMS_eff_m_id
−2
−1
0
1
2
−100
0
100
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$
Fit
Pull
+1σ Impact
-1σ Impact

HcZZ — with Z+X (true 4-era) results, 2026-09-11

Production: combine_run3_100150_ctag2d_v8_jecshifts_with_zx
[100,150] GeV, ctag2d scheme, 18 shape/lnN + 4 JES/JER systematics,
true 4-era Z+X (reducible background) included via --zx-dir.

Yields in [100,150] GeV

Process	Yield
Signal	0.0531
ggZZ	2.9418
qqZZ	36.3182
Other_Higgs	55.0776
Z+X	133.2386
Total bkg	**227.5762**

S/√B = 0.0035

Final limits (combine -M AsymptoticLimits -m 120 --run blind)

Quantile	r	κc
2.5%	219.3750	40.70
16%	311.0010	56.60
50% (median)	**480.0000**	**85.89**
84%	788.0414	139.26
97.5%	1283.8478	225.15

Comparison against related references

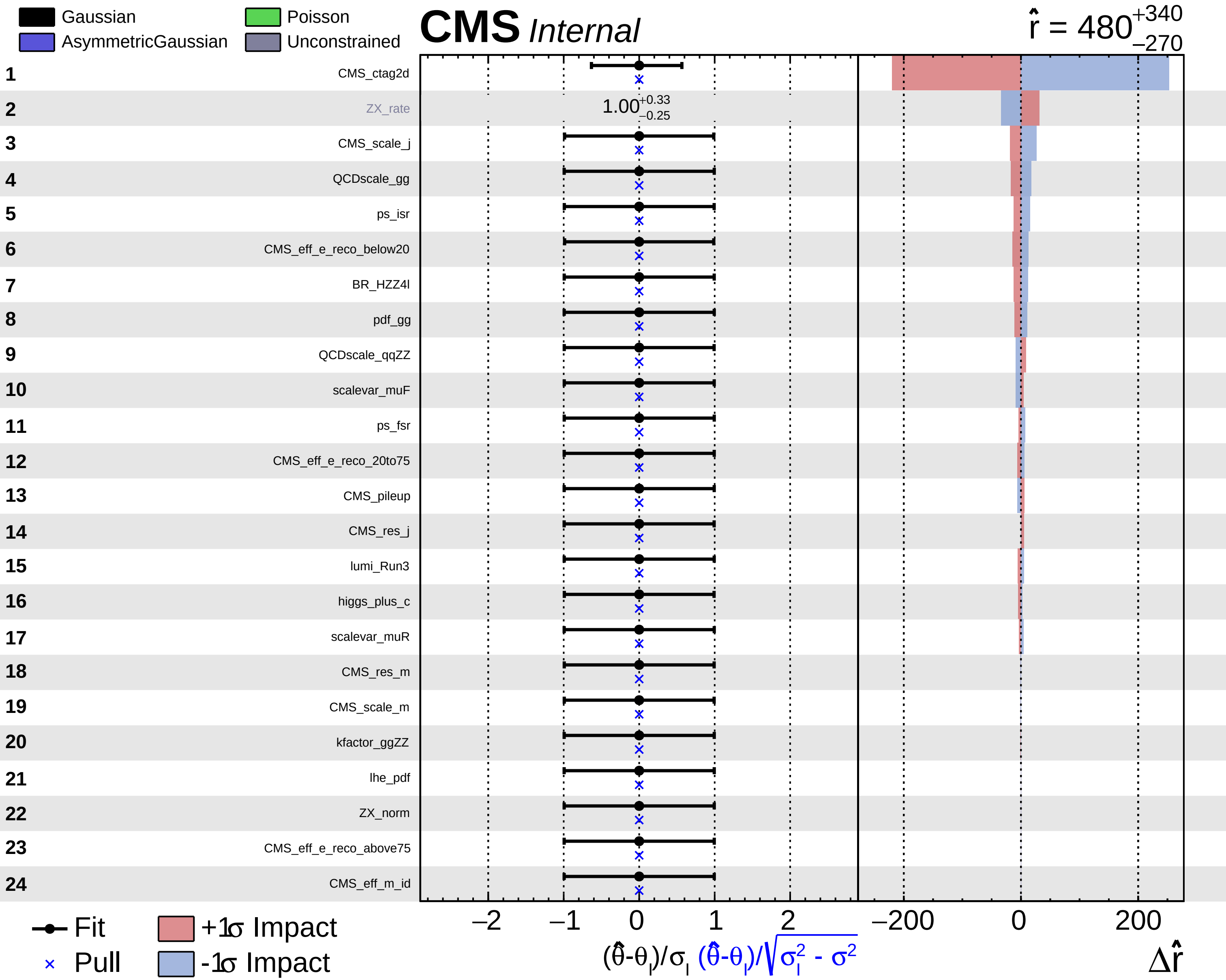
Configuration	r (median)	κc
[100,150], no Z+X, current headline (_v8+JES/JER)	412.0	74.11
[100,150], with Z+X, 2-era-only floor (older _v7, no JES/JER)	460.0	82.43
[100,150], with Z+X, true 4-era (_v8+JES/JER, current)	**480.0**	**85.89**

Result is looser than both the no-Z+X headline and the 2-era-only floor, as expected: the true 4-era Z+X yield (133.24) is ~2x the 2-era-only floor's (71.85), and more real background loosens the limit. r→κc via $\kappa_c=\sqrt{0.97r+0.03r^2}$.

Systematics with Z+X: type, assigned value, and impact

Ranked by impact_r (combineTool.py -M Impacts, --expect-signal 480.0,
true 4-era Z+X included, 24 nuisances incl. the free-floating ZX_rate).
lnN rows show the assigned size per affected process; shape rows show the
post-fit pull instead (all ≈ prefit ±1σ here, expected for a blind Asimov fit).

#	Systematic	Type	Value (lnN size / pull)	Impact
1	**CMS_ctag2d**	**shape**	**pull +0.00 (-0.633/+0.565)**	**252.66**
2	ZX_rate	rateParam	1.00 (+0.33/-0.25), free-floating	34.23
3	CMS_scale_j	shape (JES/JER)	pull +0.00 (-0.989/+0.988)	26.55
4	QCDscale_gg	lnN	1.039 (ggZZ, Signal)	17.96
5	ps_isr	shape	pull +0.00 (-0.992/+0.993)	15.28
6	CMS_eff_e_reco_below20	shape	pull +0.00 (-0.988/+0.988)	14.77
7	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	12.65
8	pdf_gg	lnN	1.032 (ggZZ, Signal)	11.01
9	QCDscale_qqZZ	lnN	1.04 (qqZZ)	8.78
10	scalevar_muF	shape	pull +0.00 (-0.993/+0.994)	8.64
11	ps_fsr	shape	pull +0.00 (-0.993/+0.994)	7.40
12	CMS_eff_e_reco_20to75	shape	pull +0.00 (-0.994/+0.994)	6.23
13	CMS_pileup	shape	pull -0.00 (-0.994/+0.994)	6.00
14	CMS_res_j	shape (JES/JER)	pull -0.00 (-0.994/+0.993)	5.30
15	lumi_Run3	lnN	1.014 (all processes)	5.28
16	higgs_plus_c	shape	pull +0.00 (-0.992/+0.992)	4.51
17	scalevar_muR	shape	pull -0.00 (-0.994/+0.994)	4.22
18	CMS_res_m	shape (JES/JER)	pull -0.00 (-0.993/+0.993)	0.87
19	CMS_scale_m	shape (JES/JER)	pull -0.00 (-0.993/+0.993)	0.77
20	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.39
21	lhe_pdf	shape	pull +0.00 (-0.994/+0.994)	0.25
22	ZX_norm	lnN	1.30 (ZX)	0.11
23	CMS_eff_e_reco_above75	shape	pull -0.00 (-0.994/+0.994)	0.05
24	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.02



Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
$\hat{r} = 480^{+340}_{-270}$
1 CMS_ctag2d
2 ZX_rate
$1.00^{+0.33}_{-0.25}$
3 CMS_scale_j
4 QCDscale_gg
5 ps_isr
6 CMS_eff_e_reco_below20
7 BR_HZZ4l
8 pdf_gg
9 QCDscale_qqZZ
10 scalevar_muF
11 ps_fsr
12 CMS_eff_e_reco_20to75
13 CMS_pileup
14 CMS_res_j
15 lumi_Run3
16 higgs_plus_c
17 scalevar_muR
18 CMS_res_m
19 CMS_scale_m
20 kfactor_ggZZ
21 lhe_pdf
22 ZX_norm
23 CMS_eff_e_reco_above75
24 CMS_eff_m_id
−2 −1 0 1 2
−200 0 200
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

Stat-only limit: with vs. without Z+X

combine -M AsymptoticLimits --freezeParameters allConstrainedNuisances
(all constrained nuisances frozen; ZX_rate still floats since it is unconstrained),
same _v8+JES/JER production, [100,150] GeV, both re-verified 2026-09-11.

Quantile	r (no ZX)	κc (no ZX)	r (with ZX)	κc (with ZX)	Δr	Δr %
2.5%	140.12	26.92	144.30	27.65	+4.18	+3.0%
16%	189.48	35.51	197.02	36.82	+7.53	+4.0%
50% (median)	**271.75**	**49.79**	**282.00**	**51.57**	**+10.25**	**+3.8%**
84%	394.69	71.11	410.14	73.79	+15.45	+3.9%
97.5%	550.66	98.14	573.54	102.10	+22.88	+4.2%

Key point: stat-only, Z+X loosens the median limit by only ~3.8% (r=271.75→282.00) — much smaller than the ~16.5% full-syst loosening (r=412.0→480.0, see the with-ZX result page above). Most of the with-ZX effect comes from systematics, not the added statistical background itself: the ZX_norm 30% lnN and the free-floating ZX_rate rateParam (impact 34.23, ranked #2 in the full-syst impact table on the previous page) account for the bulk of the gap between the stat-only and full-syst Z+X effect.